

ECE 3337: Signals & Systems Analysis

Lecture 16: Laplace Transform Applications

**Reference: Chapter 4
(Sections 4-5 & 4-6)**

**Please turn off your gadgets now,
and Focus on the Lecture**



Exam 2

Syllabus: Chapters 1 - 4, as covered in class

Time: Saturday, April 4th , 10 am – 11:30 am

Location: Science & Research Building room 117 (both sections)

Help Sessions

March 27th 10am – noon, in-person @ Cemo Hall Room 109 (Dr. Roysam)

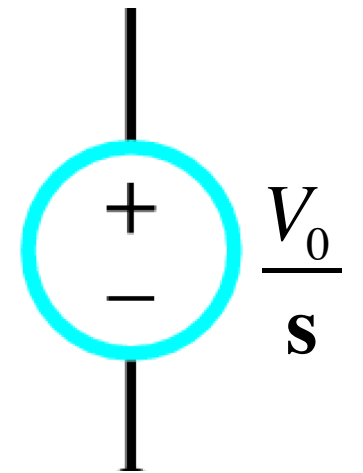
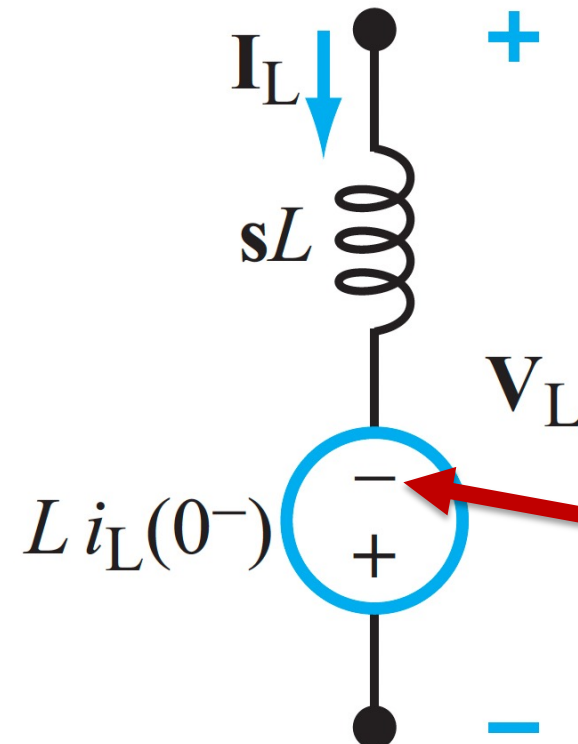
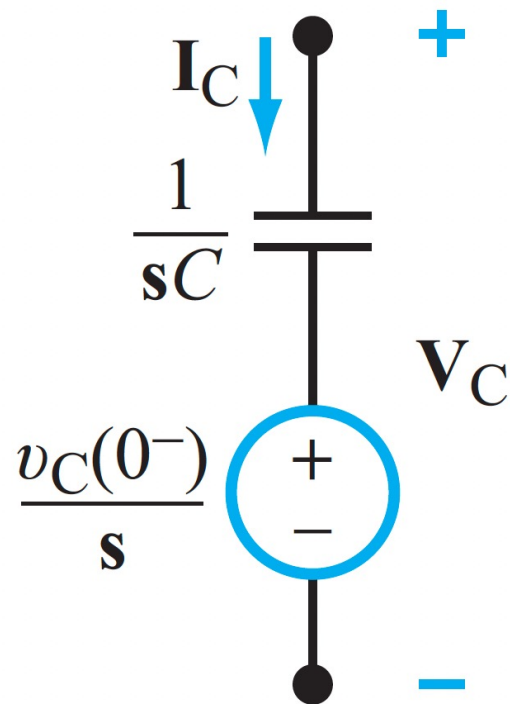
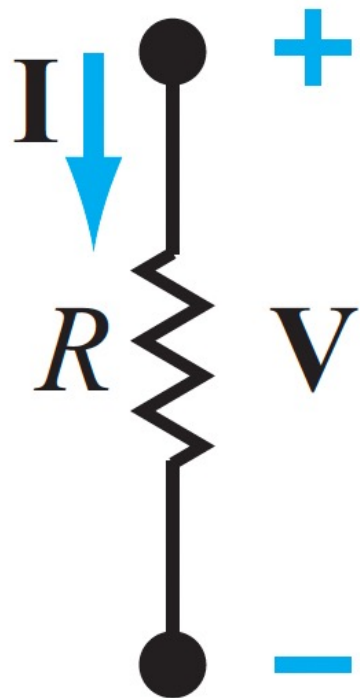
April 3rd noon – 2pm, Online via MS-TEAMS (Dr. Reddy)

Old Exams with solutions are on the Canvas site for practice

- 1. Try the old exams individually without looking at the answers**
- 2. Attempt the problems and practice until you can do them fully on your own without notes, quickly, just like in a test.**
- 3. Practice for correctness and speed**



Recap: R, L, C in the Laplace Domain



**NOTE THE BACKWARD SIGN
"REACTIVE"**



Recap: Solving Circuits with the Laplace Transform

The Laplace transform allows us to analyze complicated circuits with various initial conditions and arbitrary inputs

Step 0: Solve the circuit if necessary and determine the initial conditions

Step 1: Transform the circuit elements to the s -domain, keeping initial conditions in mind

Step 2: Transform the input signal to the s -domain

Step 3: Use KCL and KVL in the s -domain to solve for the output (parse the output into its components if needed)

Step 4: Inverse transform the output to time domain (optional)

This method is general – it applies to all kinds of circuits and signals



Today: Active Circuits

Until now, we studied passive circuits, with passive components R, L, C

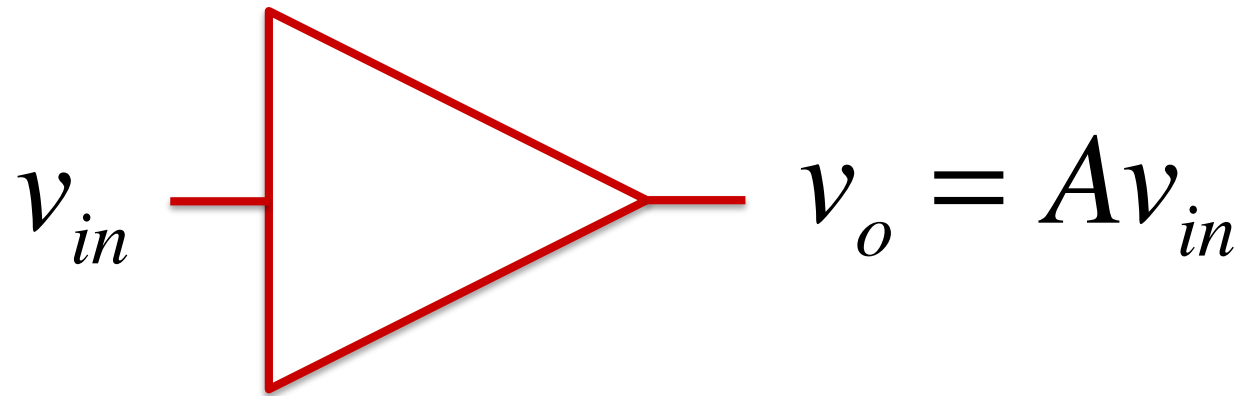
Today, we will learn about active circuits

- Active circuits contain components that need a power source (battery)
- They can inject energy into a circuit, unlike passive components
- Same analysis methods as before, but with a new “active” component: an operational amplifier (“OP AMPS”)
- OP AMPS are the most versatile and useful electronic component ever invented! An inventor’s dream come true.
- You will learn a lot more about operational amplifiers in the electronics course, so our discussion here will be less detailed, and mostly focused on applying the Laplace Transform



An Amplifier

As the name suggests, an amplifier is a device/system that can *amplify* (strengthen) a signal

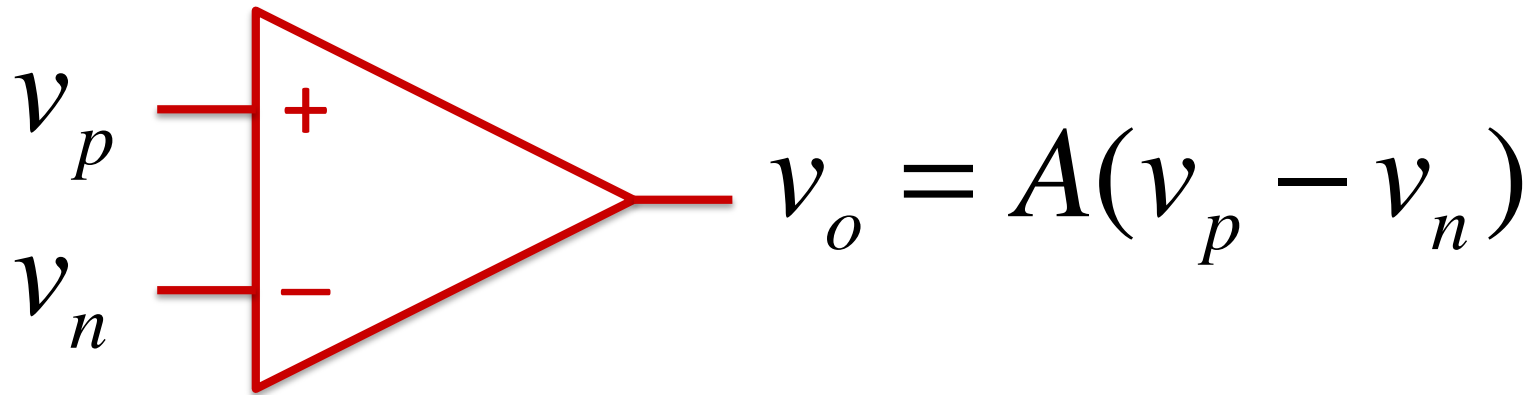


$A > 1$, and is called the
“Gain” of the amplifier

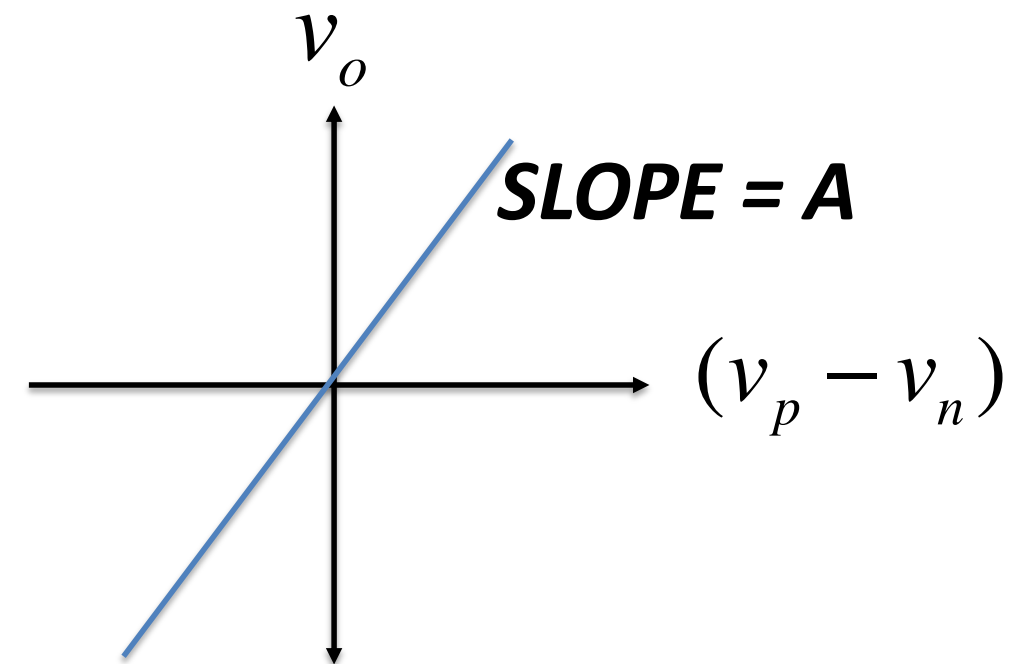


Differential Amplifier (“DIFF AMP”)

A “differential amplifier” amplifies the difference between two signals



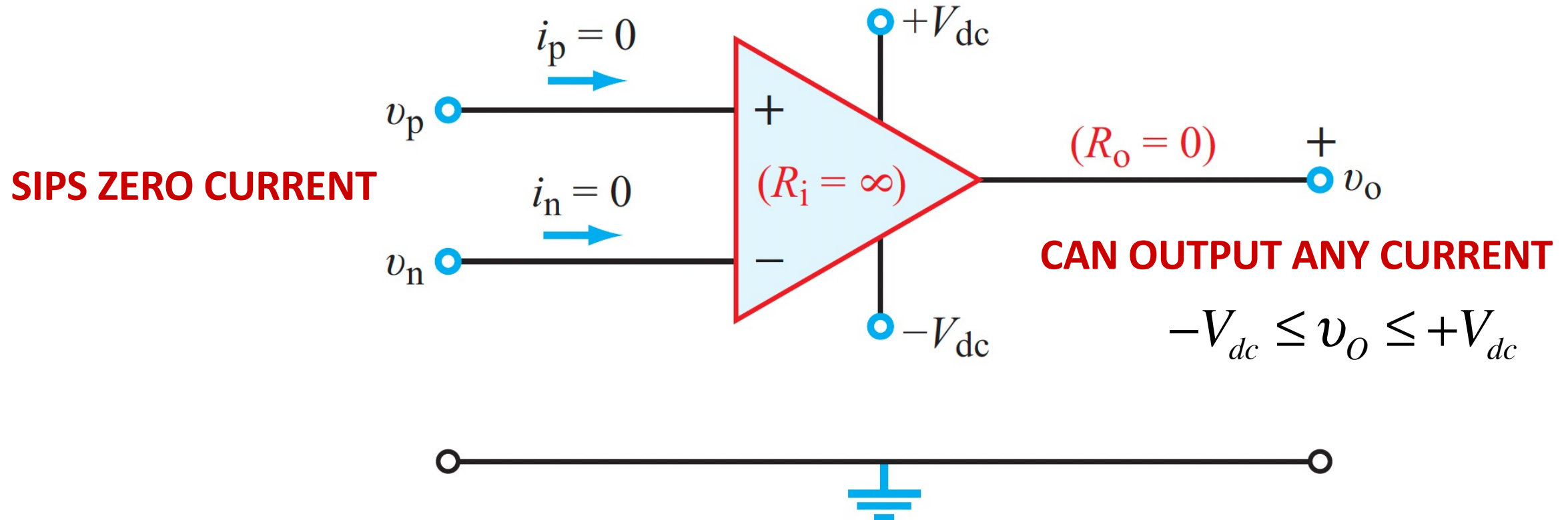
$A > 1$, and is called the
“**Gain**” of the amplifier





Operational Amplifier (“OP AMP”)

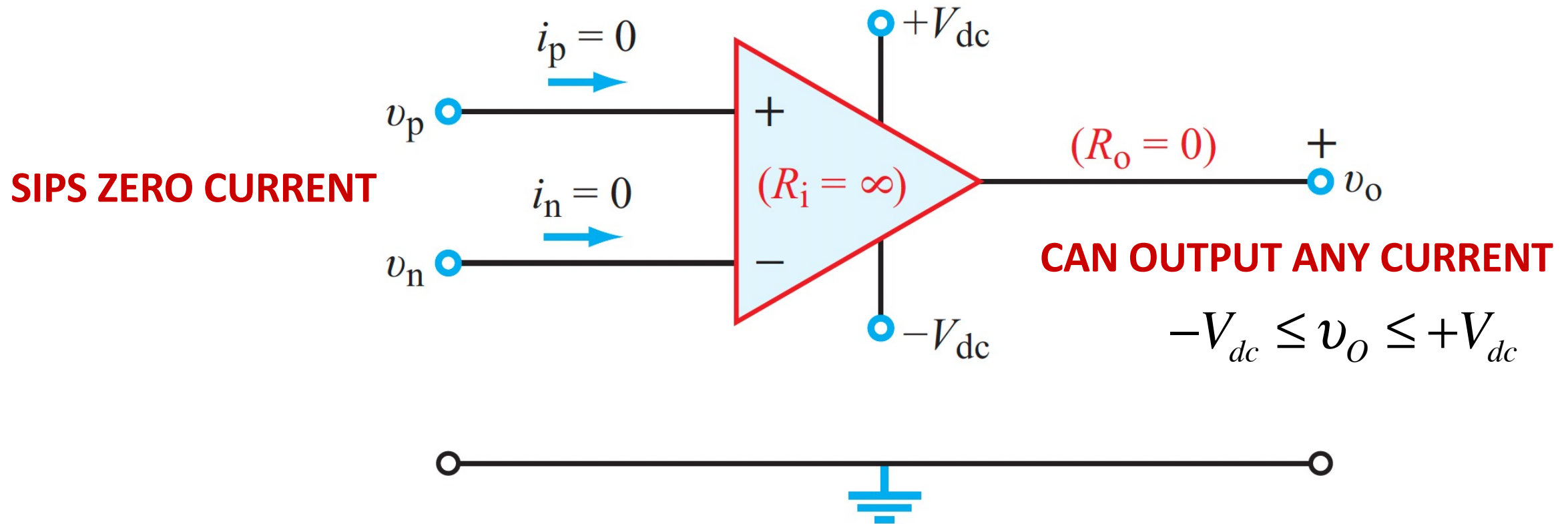
An OP AMP is an idealized “super-duper” differential amplifier with **infinite gain, zero input currents, and unlimited output current capability.**



These idealized properties make it super easy for us to design and analyze OP AMP circuits, as we will see



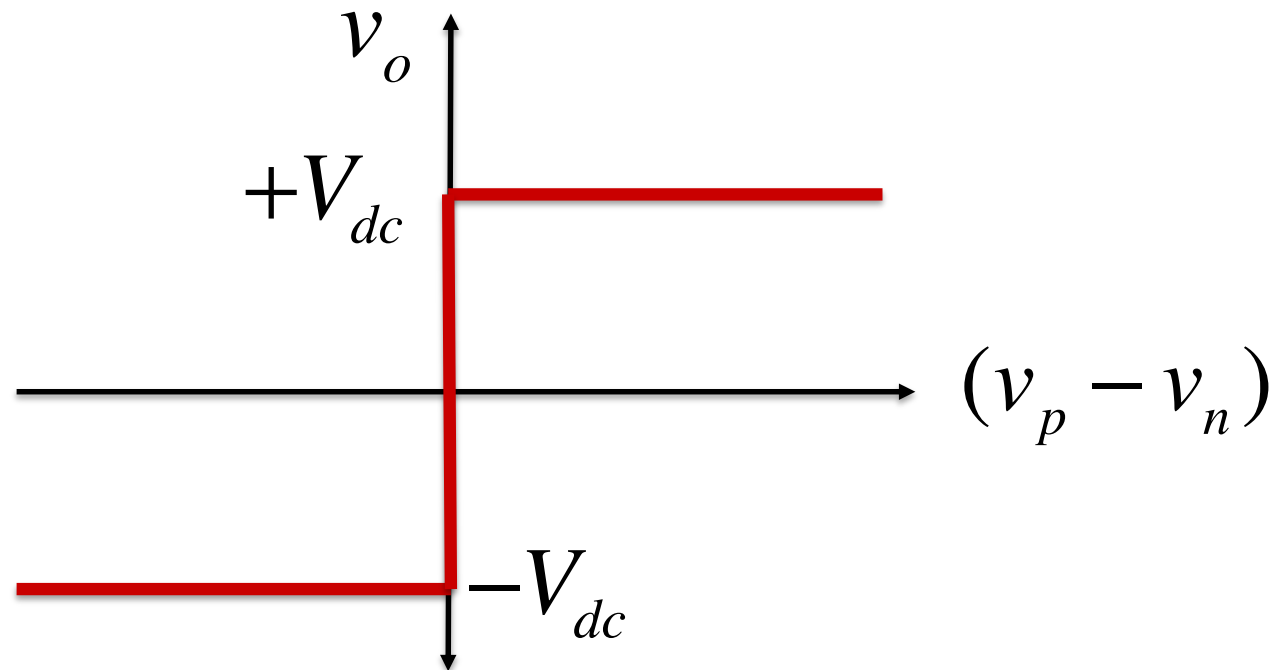
Operational Amplifier (“OP AMP”)



Zero input currents effectively make the input impedance R_i infinite
Ability to output any current make the output impedance R_o zero



Limits on OP AMP Output Voltage

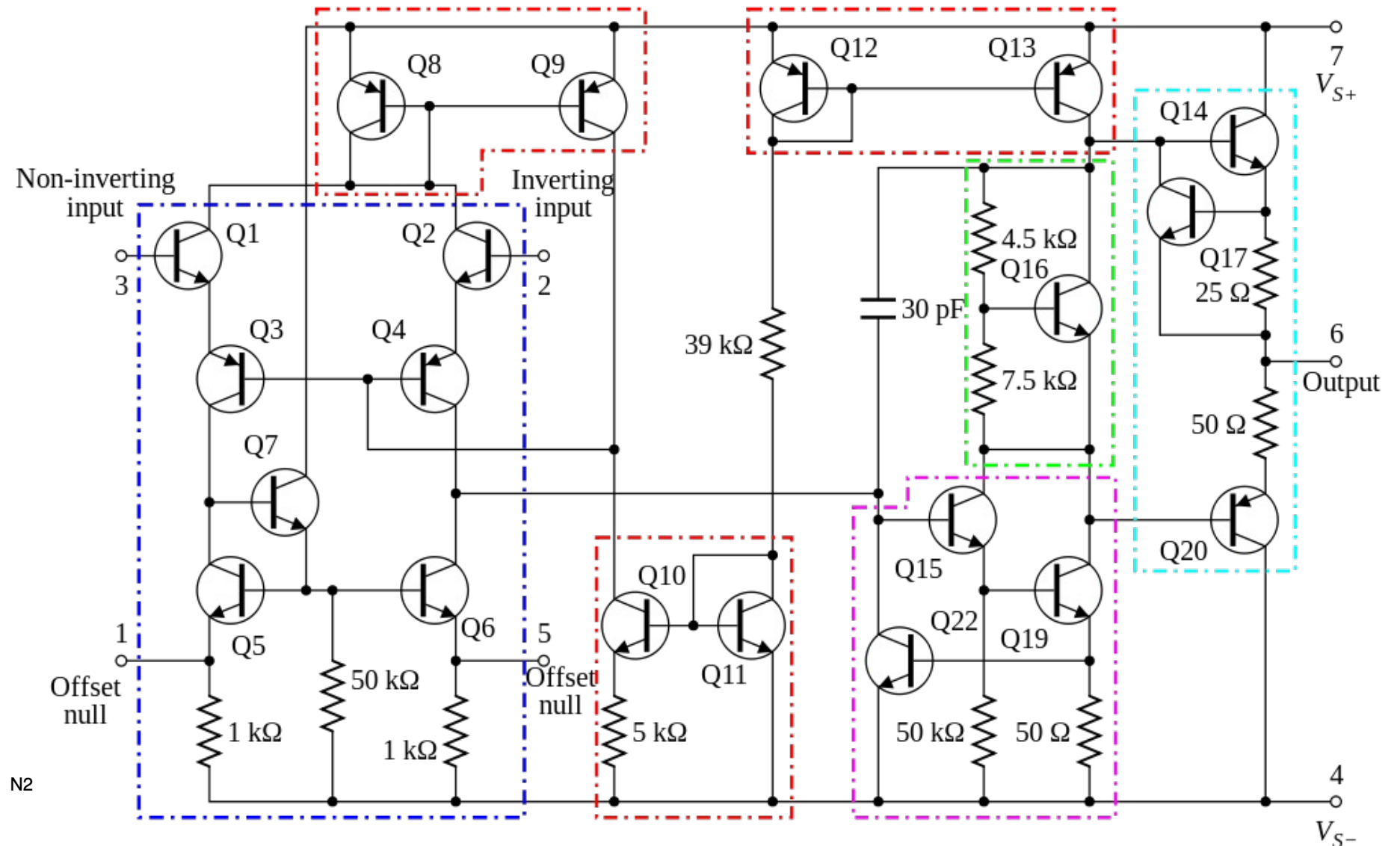
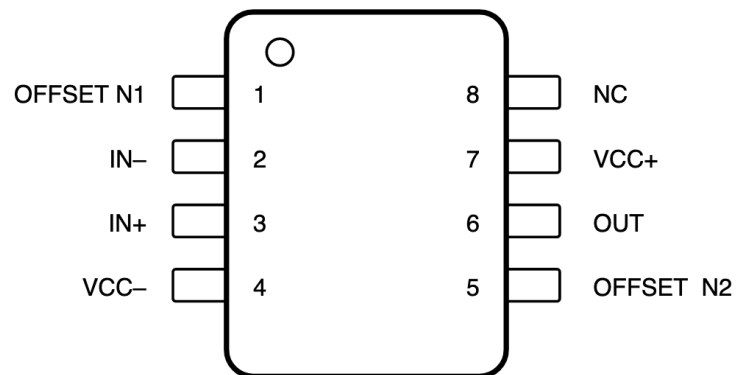


- The slope is infinite, since the gain (A) is infinite
- The output voltage is fundamentally limited by the power supply voltages

$$-V_{dc} \quad \dots \quad +V_{dc}$$



Peek Inside the op-amp chip

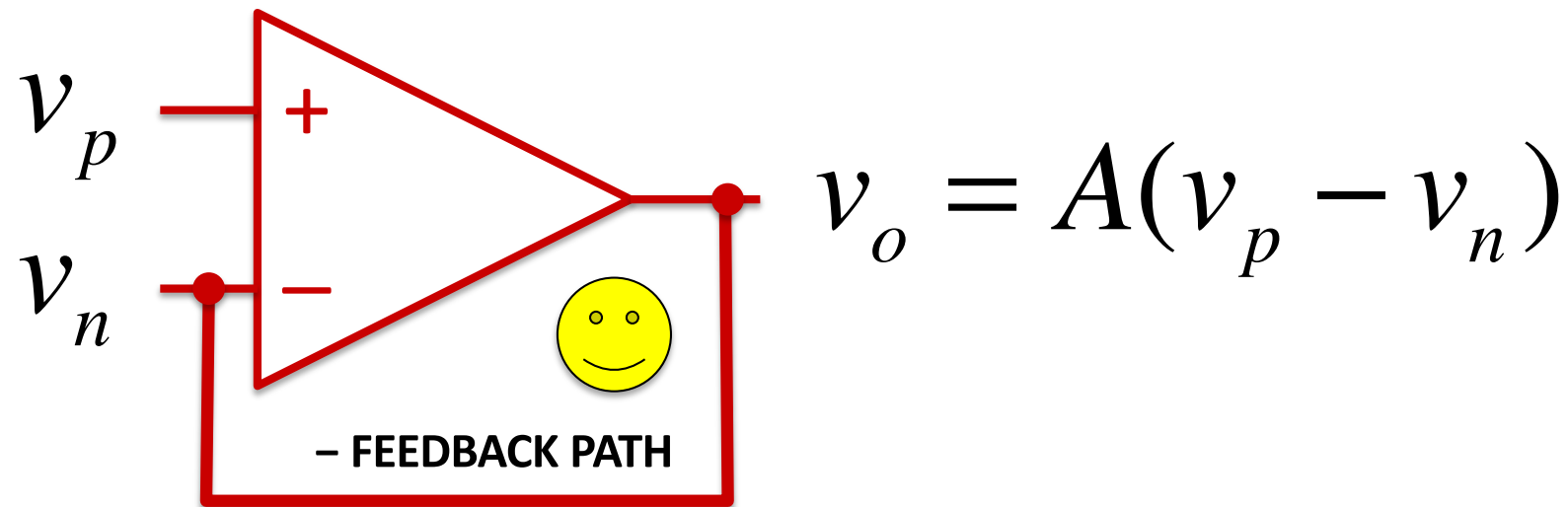




Op Amp in Stable Mode

In this course, we will only concern ourselves with Op Amps that are in stable mode:

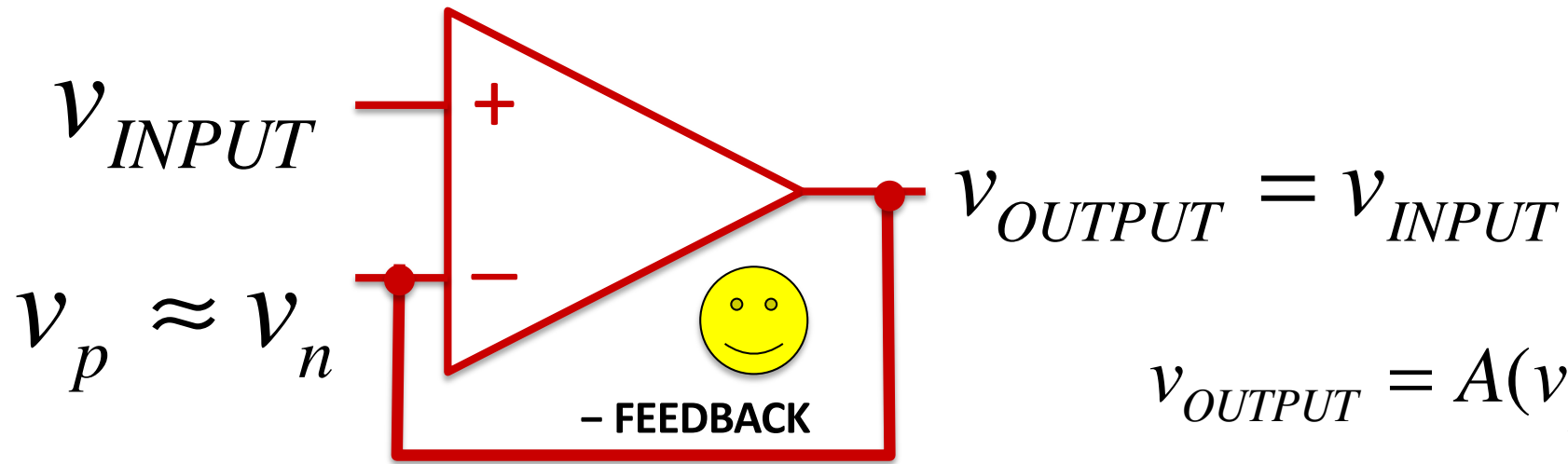
- There is a DC path (resistor/inductor) connecting the output to the negative input terminal: this is *negative feedback*.



- Intuitively, small increases in the output voltage will work to counter-act the increase



Voltage Follower



$$v_{OUTPUT} = A(v_p - v_n) = A(v_p - v_{OUTPUT})$$

$$\Rightarrow v_{OUTPUT} = \frac{A}{(1 + A)} v_p = \frac{A}{(1 + A)} v_{INPUT}$$

$$\lim_{A \rightarrow \infty} \{ v_{OUTPUT} \} = v_{INPUT}$$

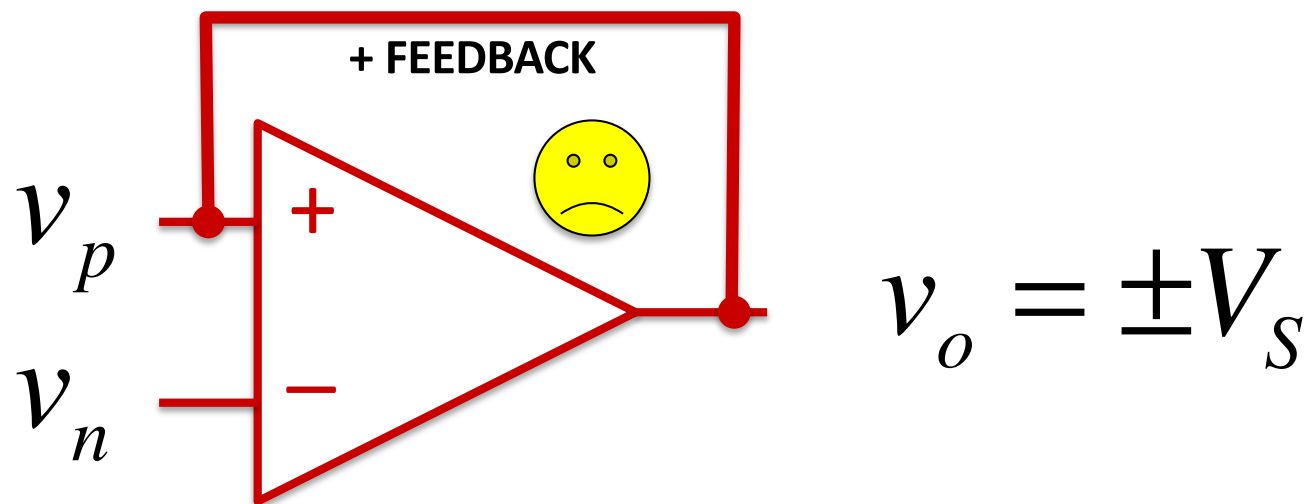
$$\lim_{A \rightarrow \infty} \{ v_p - v_n \} = 0$$



Op Amp in Unstable Mode

If we were to connect the output to the positive input instead, we get an unstable situation!

- There is a dc path (resistor/inductor) connecting the output to the positive input terminal



- Intuitively, small increases in the output voltage will work to further accentuate the increase, until the practical limits are reached!



Precise Amplification of Signals

The gain of an op-amp is only guaranteed to be greater than a threshold, for example, $A > 50,000$

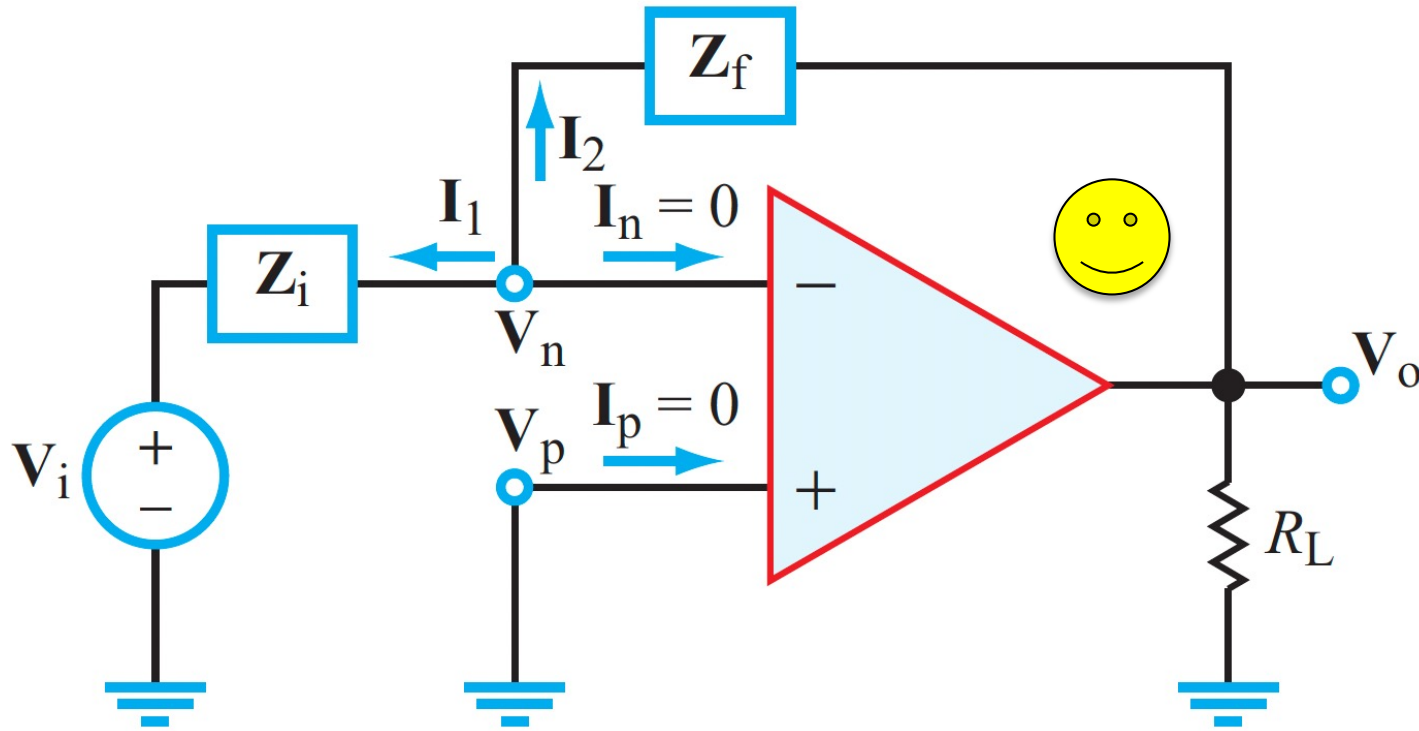
- The actual value is not precisely set
- Can vary quite a bit!

Usually, we need to be able to amplify a signal by an accurate amount, say 5.23,

- This can be achieved using variations of the voltage follower
- The idea is to specify the gain by values of resistors, and this can be made super accurate!



Inverting Amplifier



With negative feedback established, if allowed by Z_f ,
 $V_p = V_n$

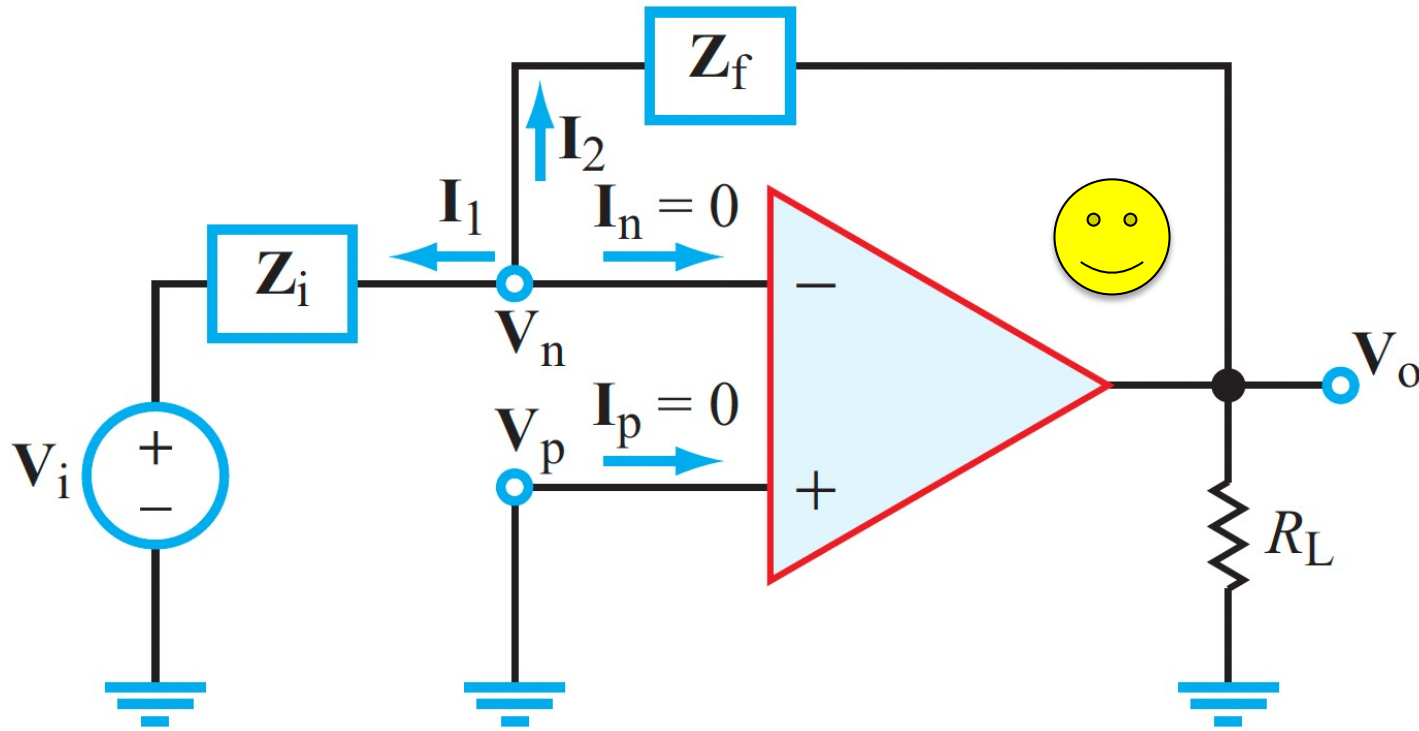
$$\cancel{I_1} + I_2 + \cancel{I_n} = 0$$

$$V_p = V_n = 0$$

$$\cancel{\frac{V_n - V_i}{Z_i}} + \cancel{\frac{V_n - V_o}{Z_f}} + \cancel{I_n} = 0$$

$$\Rightarrow V_o(s) = -\frac{Z_f(s)}{Z_i(s)} V_i(s)$$

Special Case 1



$$Z_f(s) = R_f$$

$$Z_i(s) = R_i$$

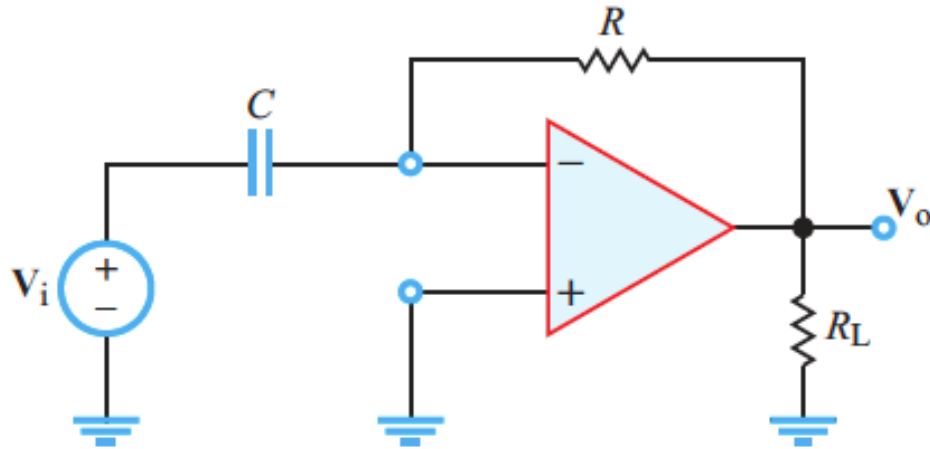
Choose the resistors to achieve the desired signal amplification

$$\Rightarrow V_o(s) = -\frac{R_f}{R_i} V_i(s)$$

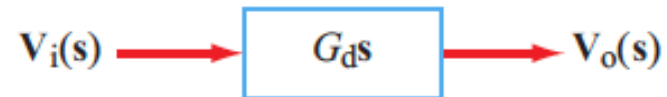
Independent of the OP-AMP's characteristics



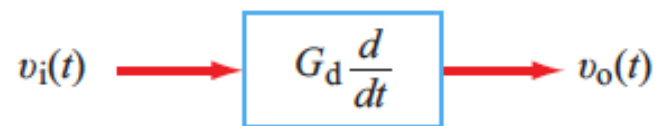
Special Case 2



$$G_d = -RC$$



s-domain



Time domain

$$Z_f(s) = R$$

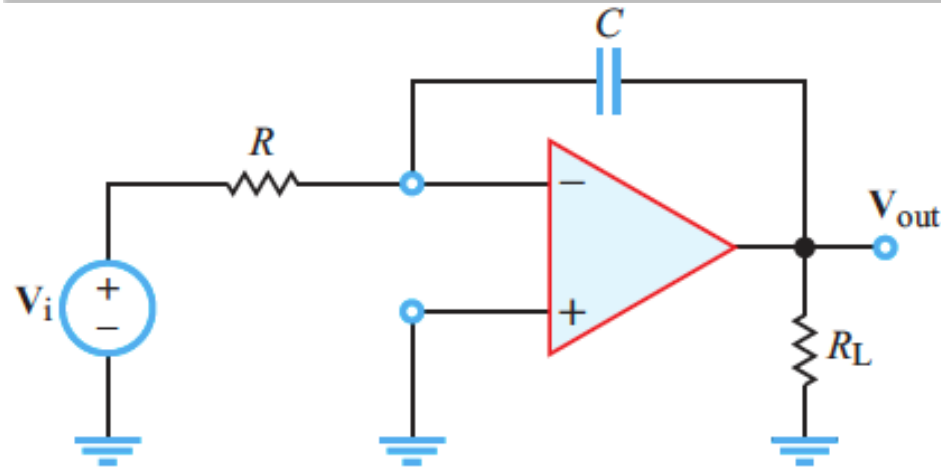
$$Z_i(s) = 1 / sC$$

$$\Rightarrow V_o(s) = -\frac{R}{1/sC} V_i(s) = -RCs V_i(s)$$

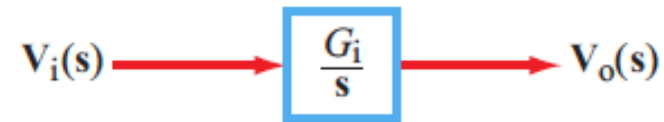
Signal Differentiator



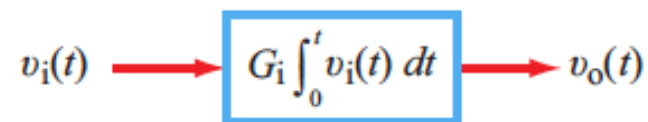
Special Case 3



$$G_i = -1/RC$$



s-domain



Time domain

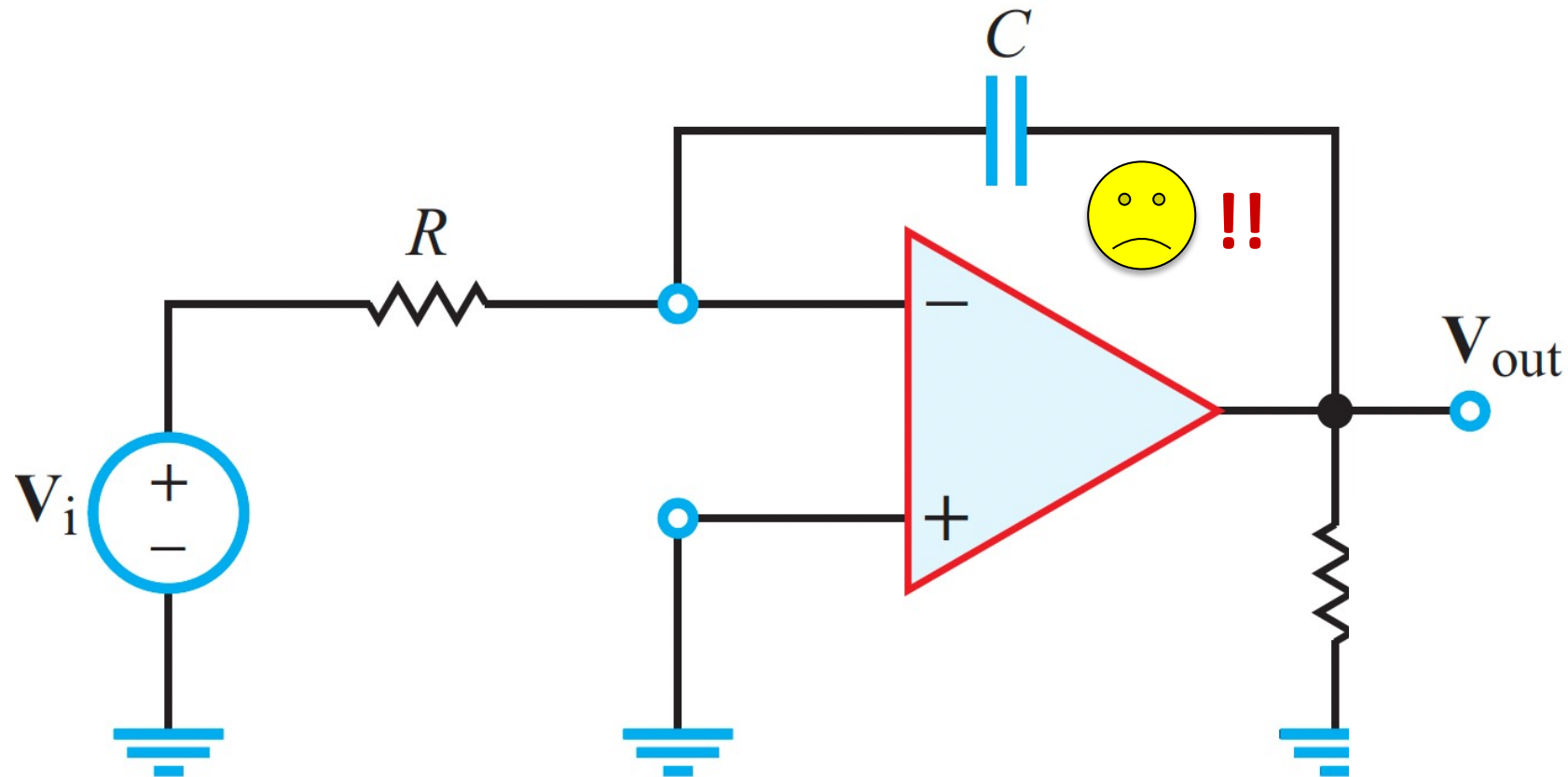
$$\mathbf{Z}_f(s) = 1 / sC$$

$$\mathbf{Z}_i(s) = R$$

$$\Rightarrow \mathbf{V}_o(s) = -\frac{1 / sC}{R} \mathbf{V}_i(s) = -\frac{1}{sRC} \mathbf{V}_i(s)$$

Signal Integrator

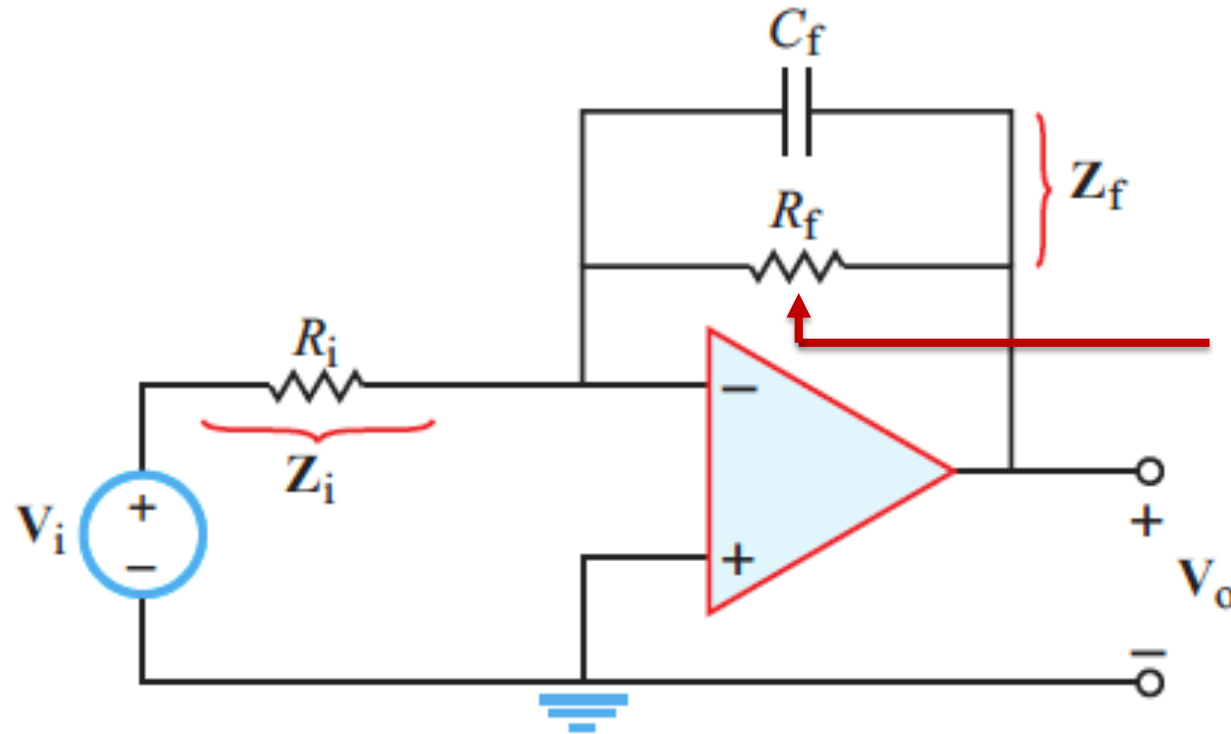
Practical Note



**The Integrator is not in dc stable mode
(what can we do to make it stable?)**



Example Circuit



**LEAKAGE RESISTOR
TO ENSURE DC
NEGATIVE FEEDBACK**

$$V_o(s) = -\frac{R_f \parallel (1/sC_f)}{R_i} V_i(s)$$

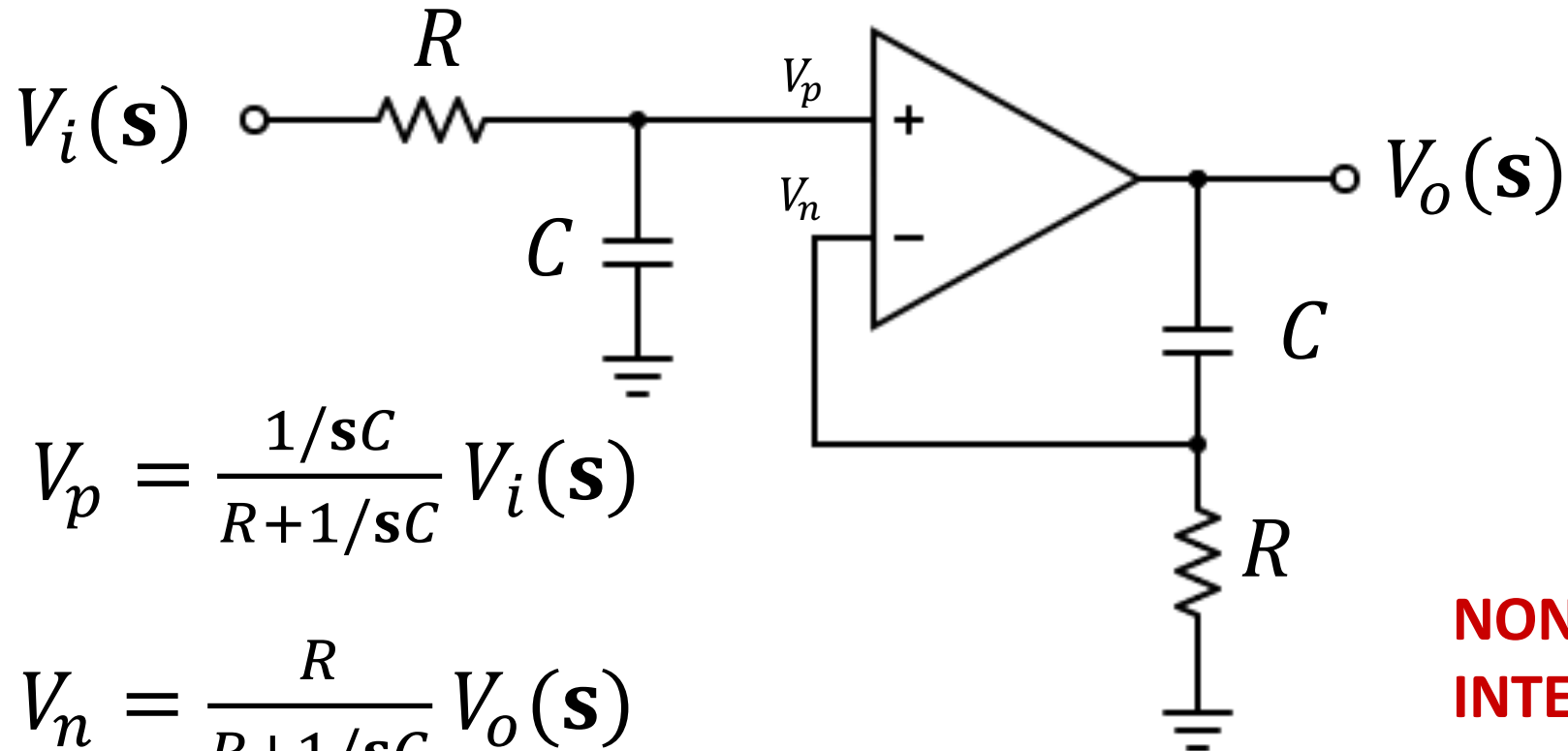
When R_f is very large, so $s + \frac{1}{R_f C_f} \approx s$, this is a close approximation to an integrator.

$$\frac{V_o(s)}{V_i(s)} = -\left(\frac{1}{R_i C_f}\right) \left[\frac{1}{s + 1/(R_f C_f)} \right]$$

For non-extreme values of R_f , this circuit behaves as a single-pole system



Non-Inverting Integrator



$$V_p = \frac{1/sC}{R+1/sC} V_i(s)$$

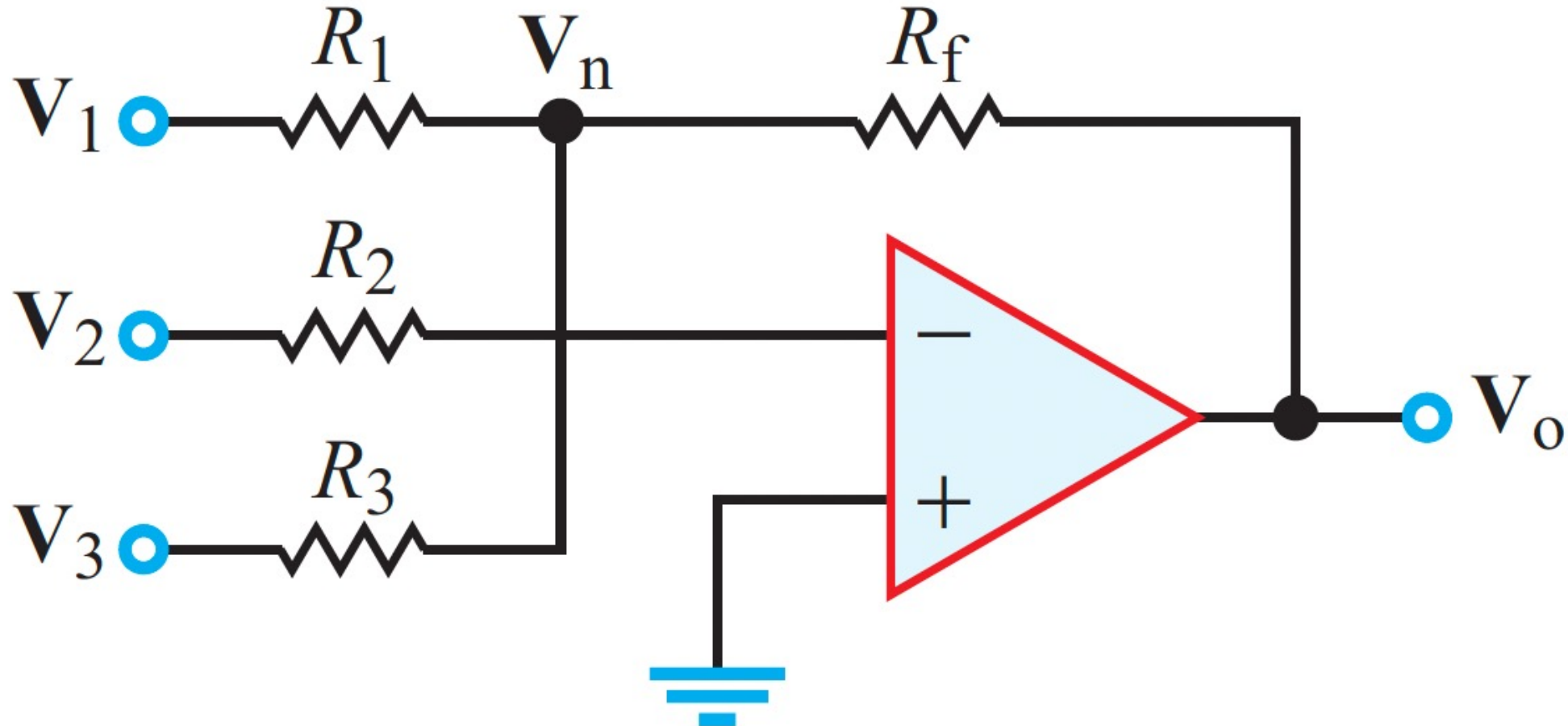
$$V_n = \frac{R}{R+1/sC} V_o(s)$$

**NON-INVERTING
INTEGRATOR**

$$\begin{aligned} V_p &= V_n \\ \Rightarrow \frac{1/sC}{R+1/sC} V_i(s) &= \frac{R}{R+1/sC} V_o(s) \end{aligned} \quad \Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{1}{sRC}$$



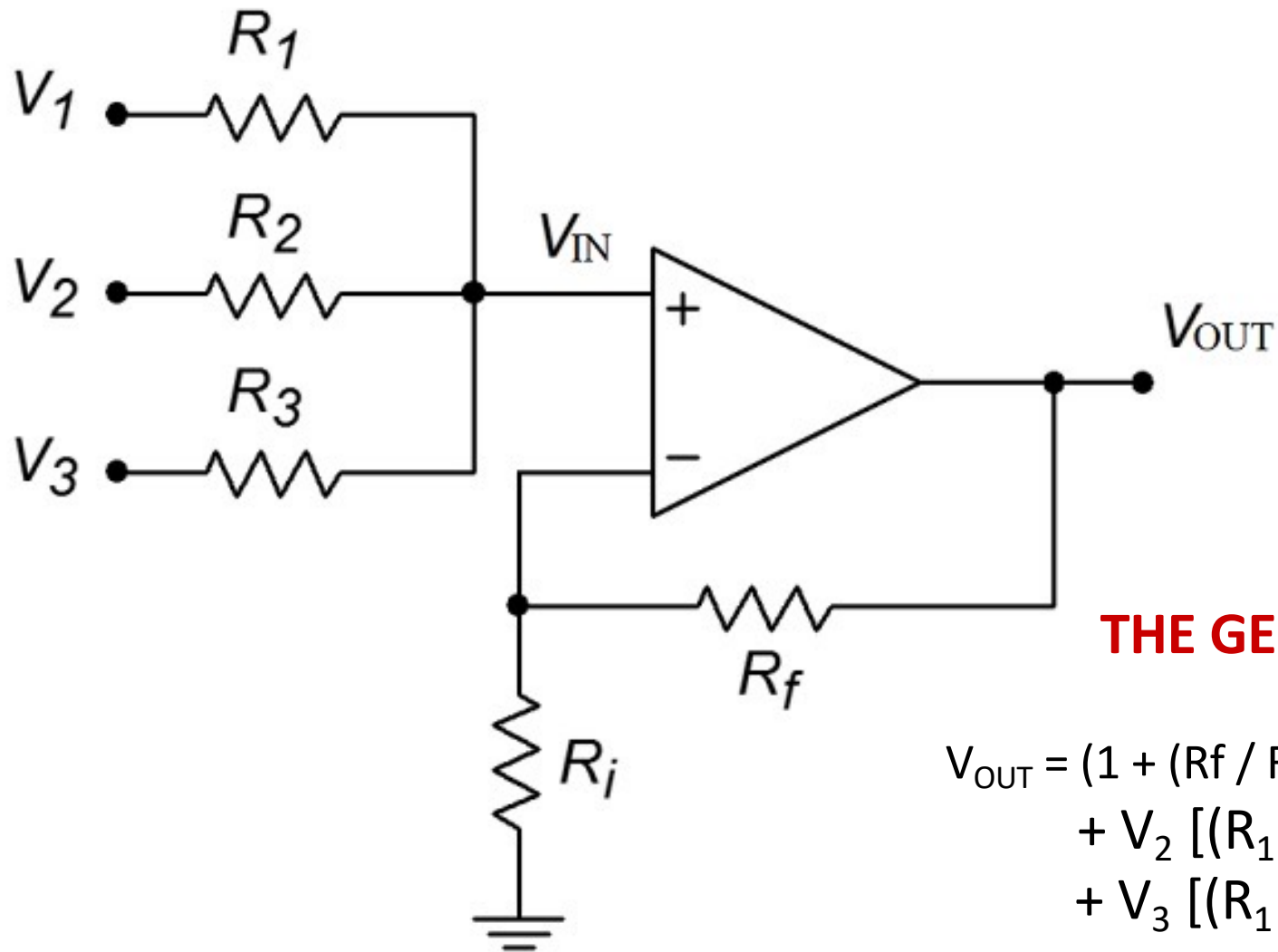
Inverting Adder (Summer)



$$\mathbf{V}_o(s) = -R_f \left[\frac{\mathbf{V}_1(s)}{R_1} + \frac{\mathbf{V}_2(s)}{R_2} + \frac{\mathbf{V}_3(s)}{R_3} \right]$$



Non-inverting Adder (Summer)



WHEN ALL RESISTORS ARE = R ,

$$V_{OUT} = \frac{2}{3} (V_1 + V_2 + V_3)$$

THE GENERAL FORMULA IS MESSY

$$\begin{aligned} V_{OUT} = & (1 + (R_f / R_i)) \{ V_1 [(R_2 || R_3) / (R_1 + (R_2 || R_3))] \\ & + V_2 [(R_1 || R_3) / (R_2 + (R_1 || R_3))] \\ & + V_3 [(R_1 || R_2) / (R_3 + (R_1 || R_2))] \} \end{aligned}$$



Adders Rock!





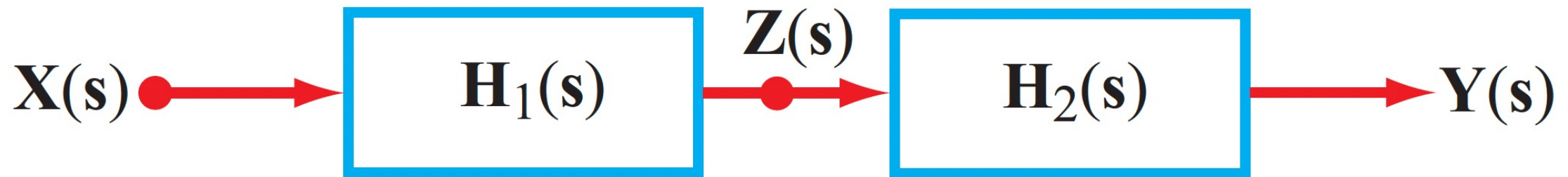
Adders Rock!



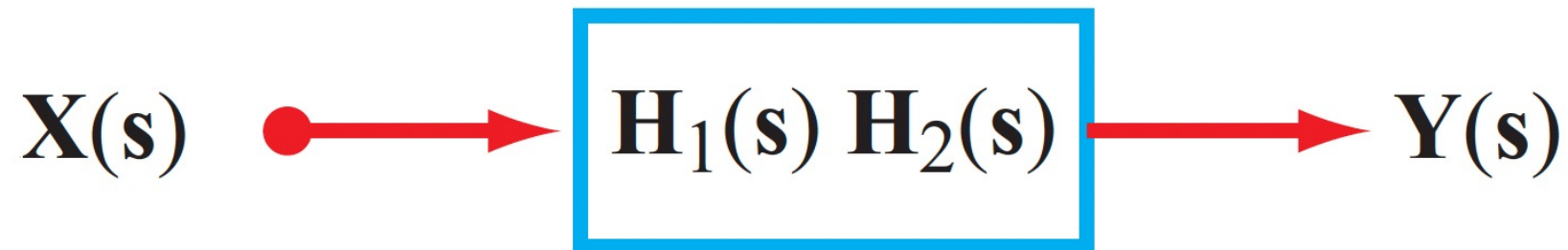


Cascading Systems

Complex systems are often built by connecting simpler systems together



Equivalent System:

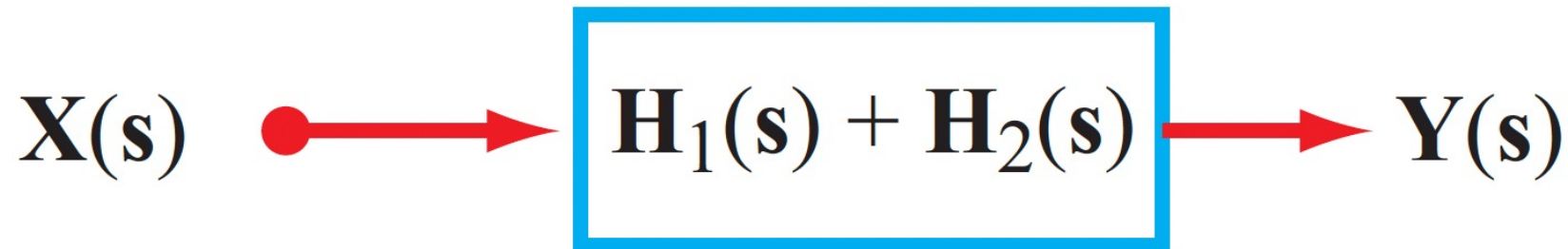
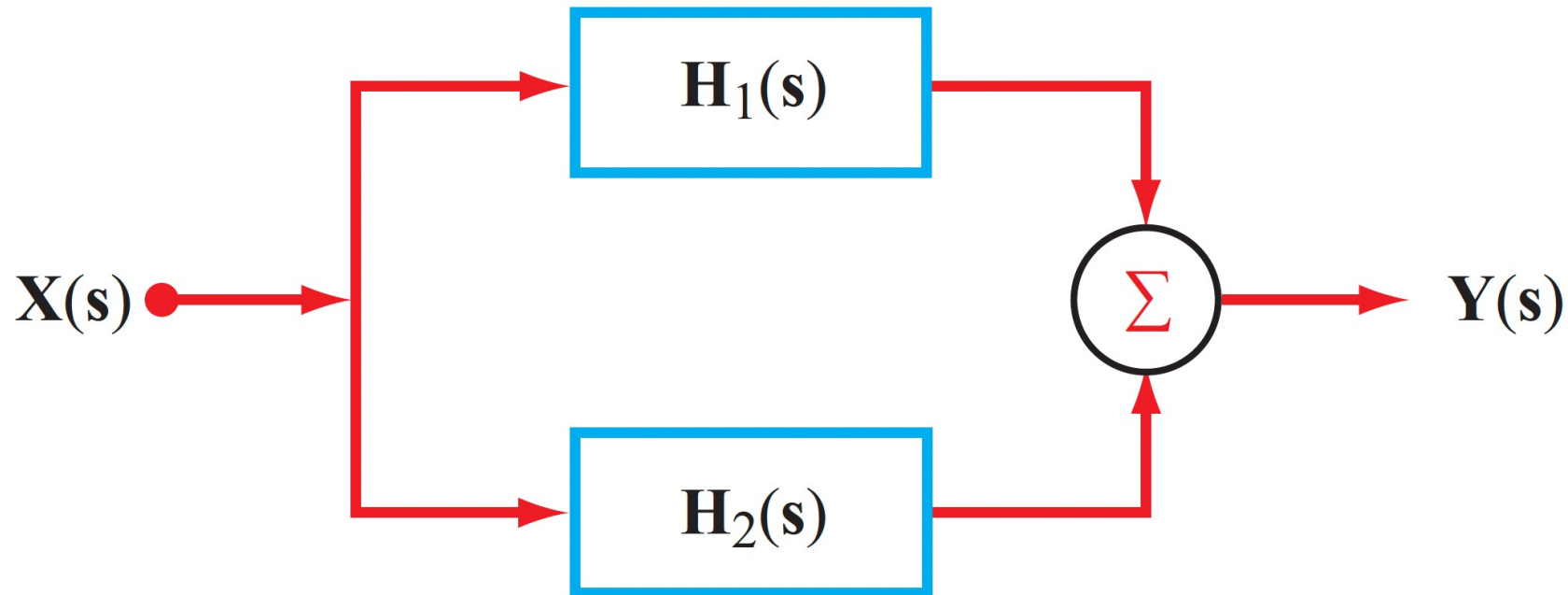




Parallel Systems

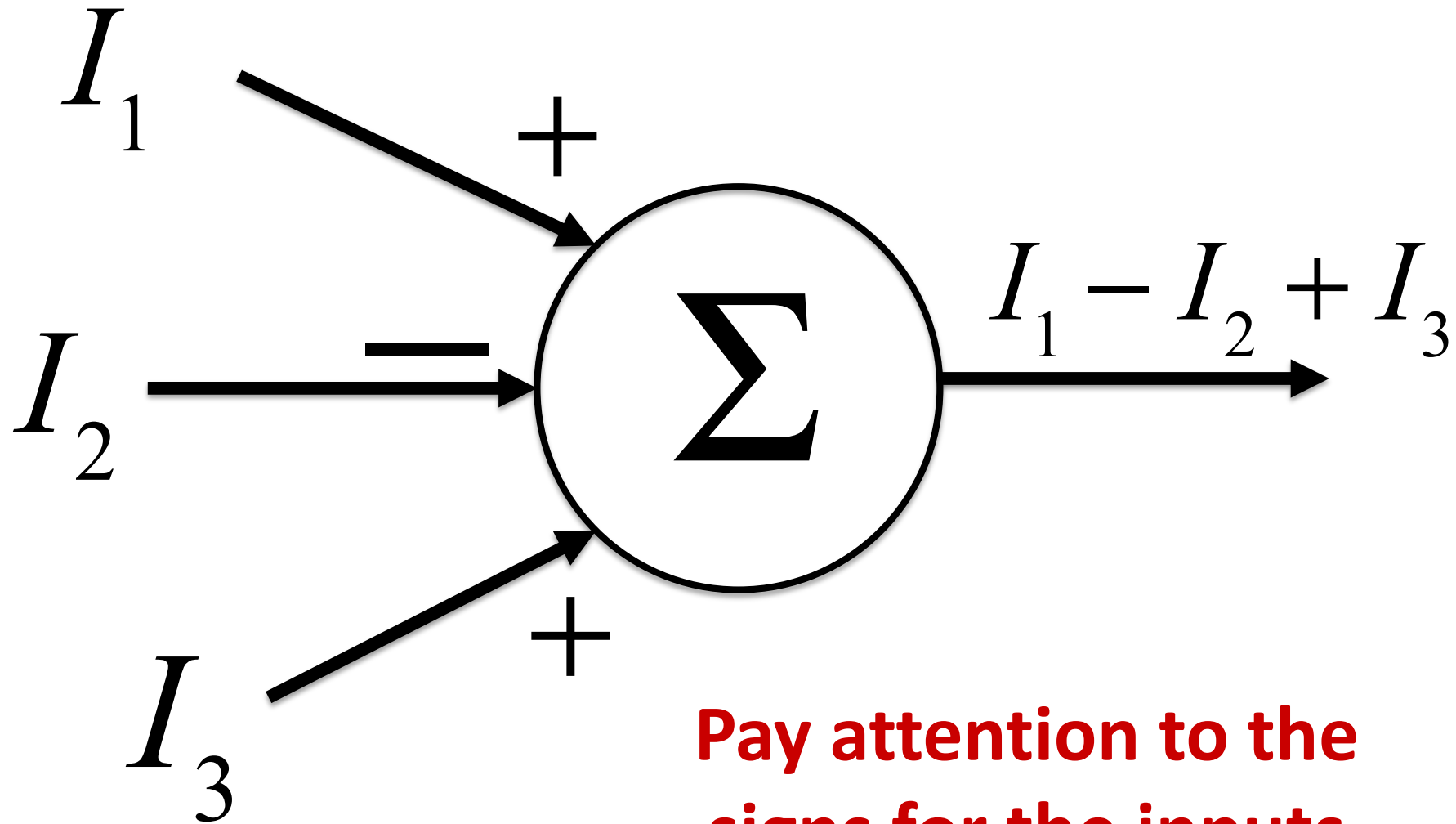
We can also add up the outputs of multiple systems

– Can you think of an application to music?





Adder Notation

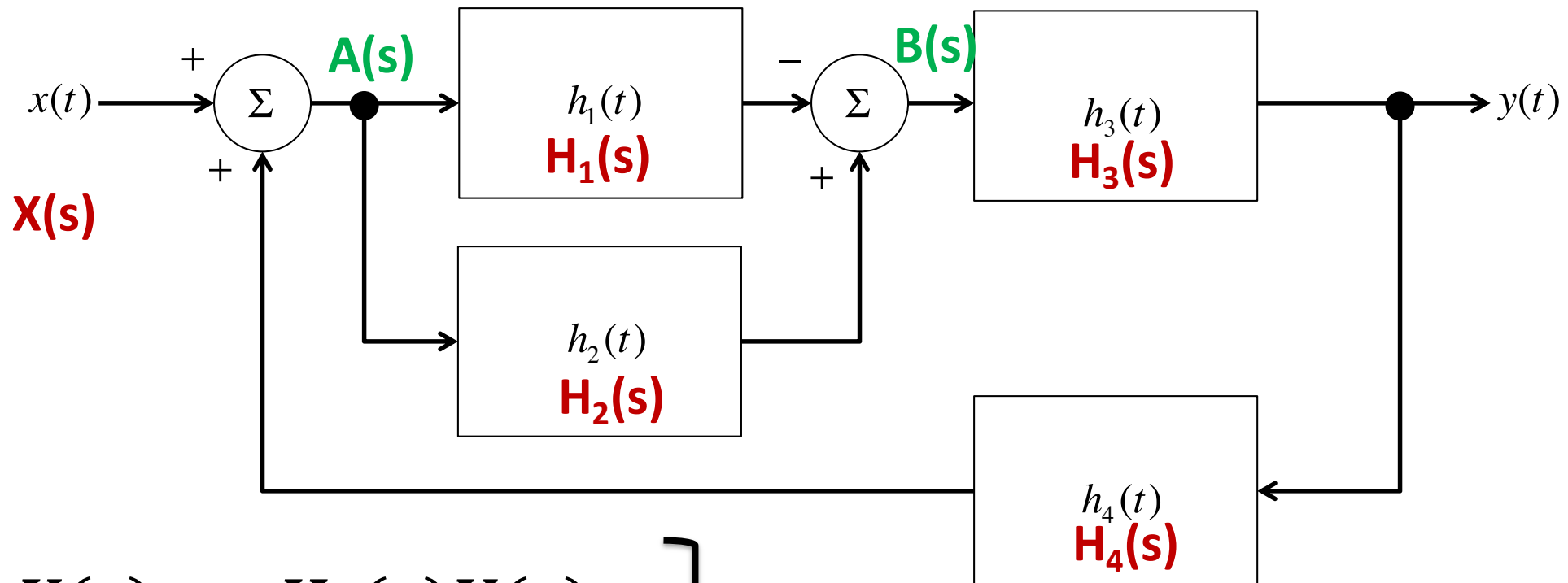


Pay attention to the signs for the inputs



Example: Solving Block Diagrams

Derive the transfer function of the following system



$$A(s) = X(s) + H_4(s)Y(s)$$

$$B(s) = A(s)(H_2(s) - H_1(s))$$

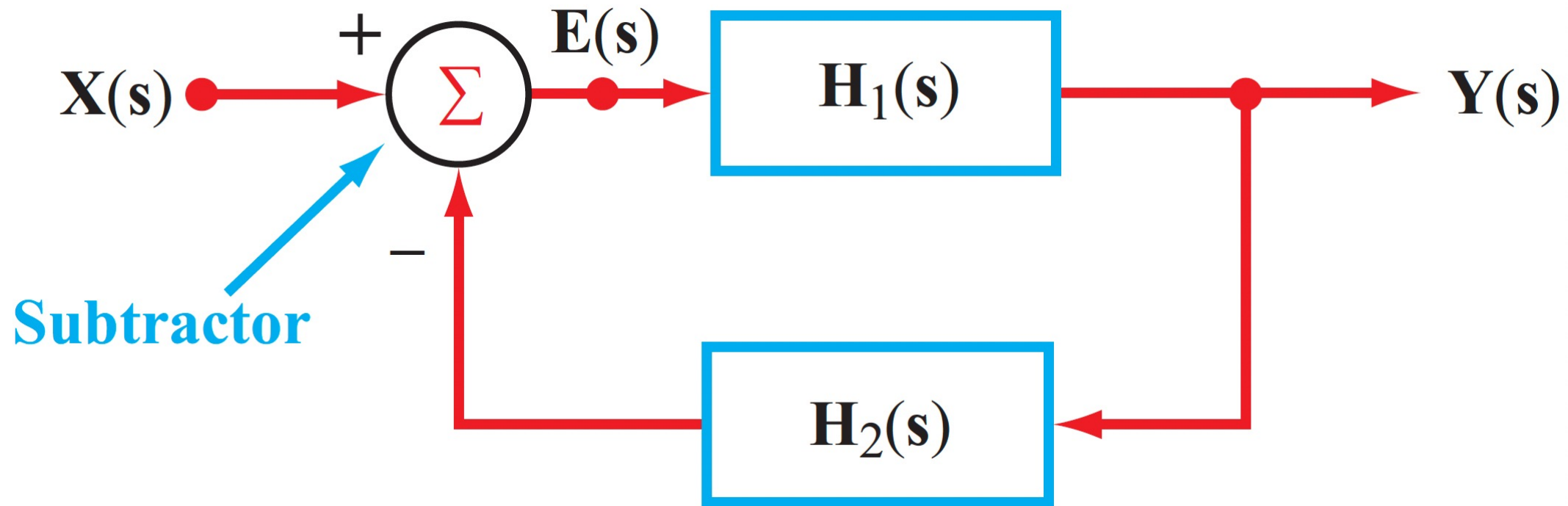
$$Y(s) = H_3(s)B(s)$$

$$\left. \begin{array}{l} A(s) = X(s) + H_4(s)Y(s) \\ B(s) = A(s)(H_2(s) - H_1(s)) \\ Y(s) = H_3(s)B(s) \end{array} \right\} \Rightarrow H(s) = \frac{Y(s)}{X(s)} = \frac{H_3(H_2 - H_1)}{1 - H_3H_4(H_2 - H_1)}$$



Feedback Connection

When we feed back the output to the input, truly interesting things can happen



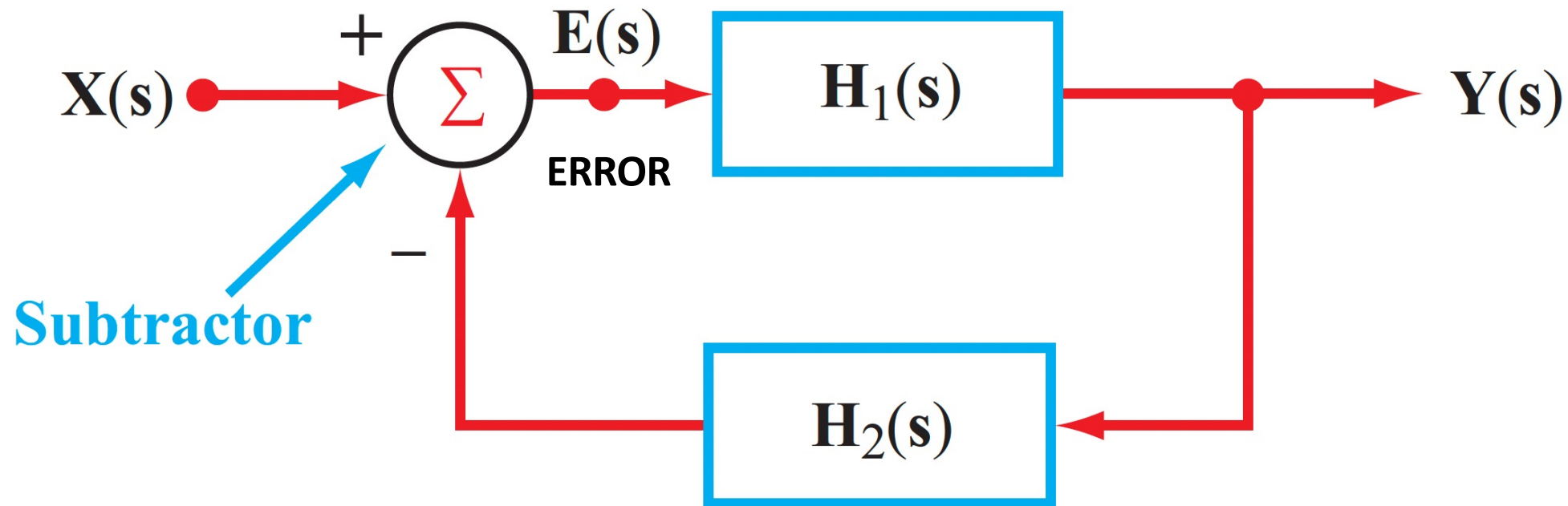
This type of circuit is used in control systems

– There is a separate course for that



Negative Feedback Connection

Let's analyze this system using Laplace Transforms



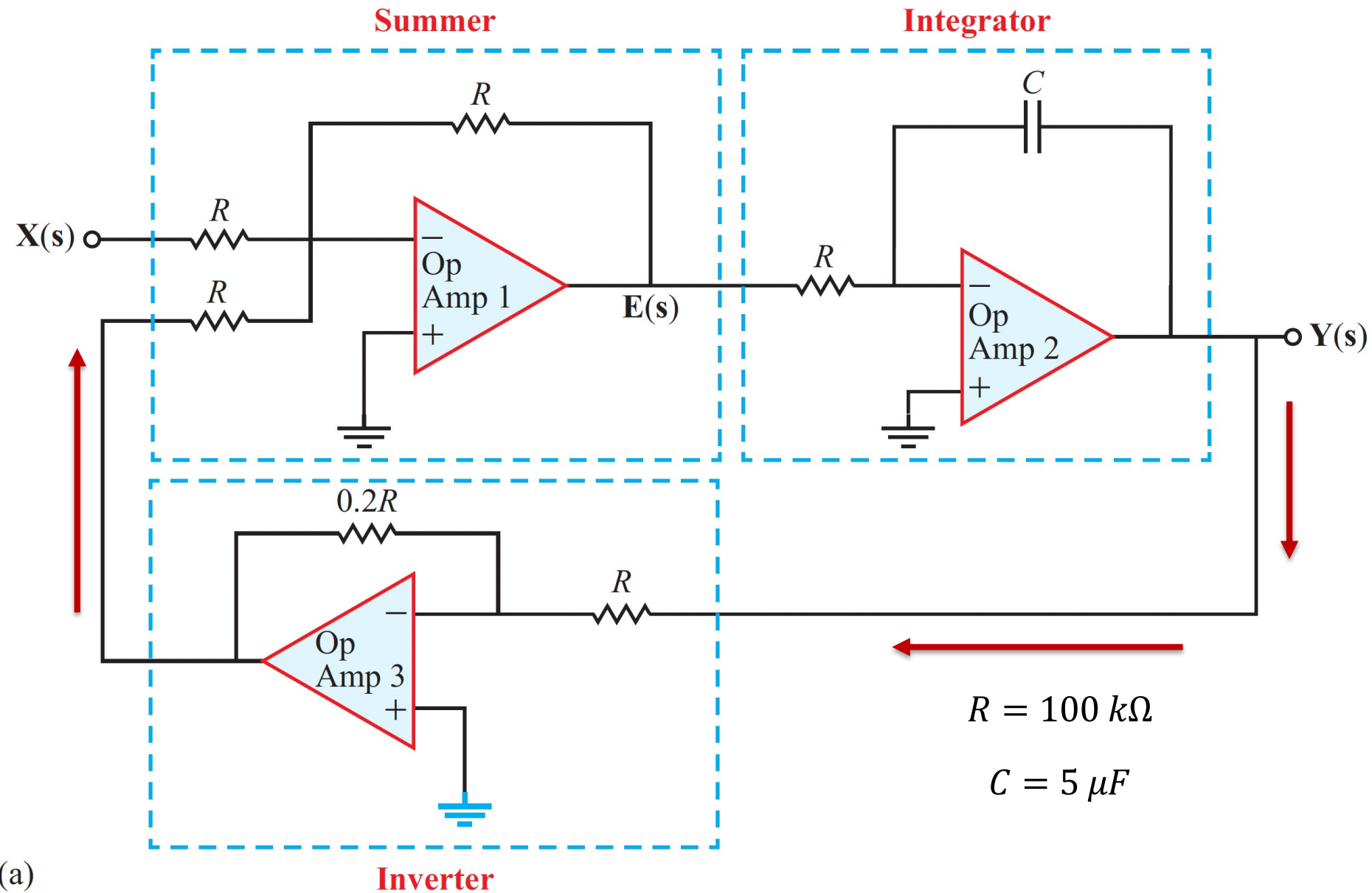
$$E(s) = X(s) - Y(s)H_2(s)$$

$$Y(s) = E(s)H_1(s)$$

$$\Rightarrow H(s) = \frac{Y(s)}{X(s)} = \frac{H_1(s)}{1 + H_1(s)H_2(s)}$$



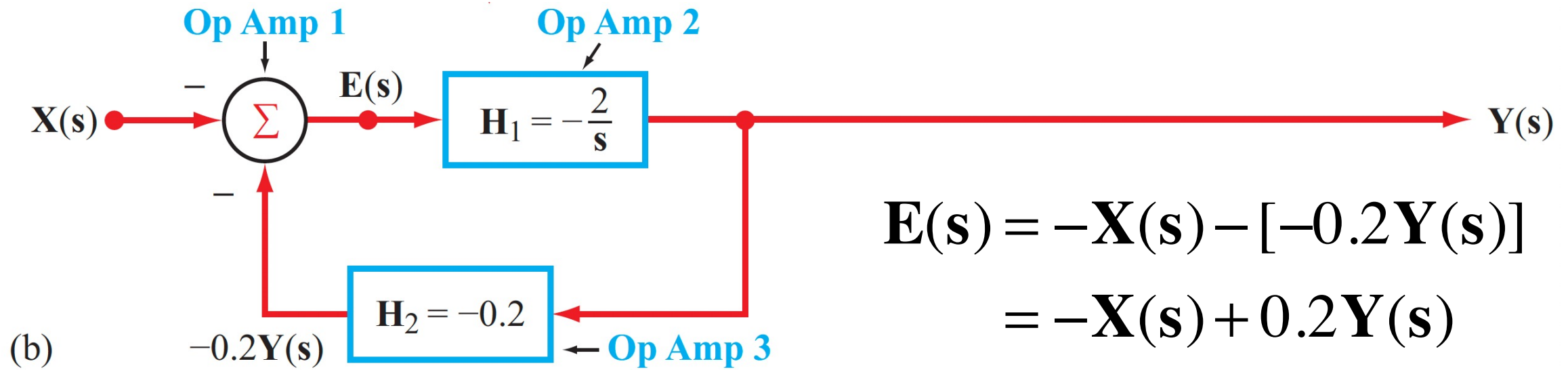
Op Amp Example



(a)



Equivalent Block Diagram



$$\begin{aligned} E(s) &= -X(s) - [-0.2Y(s)] \\ &= -X(s) + 0.2Y(s) \end{aligned}$$

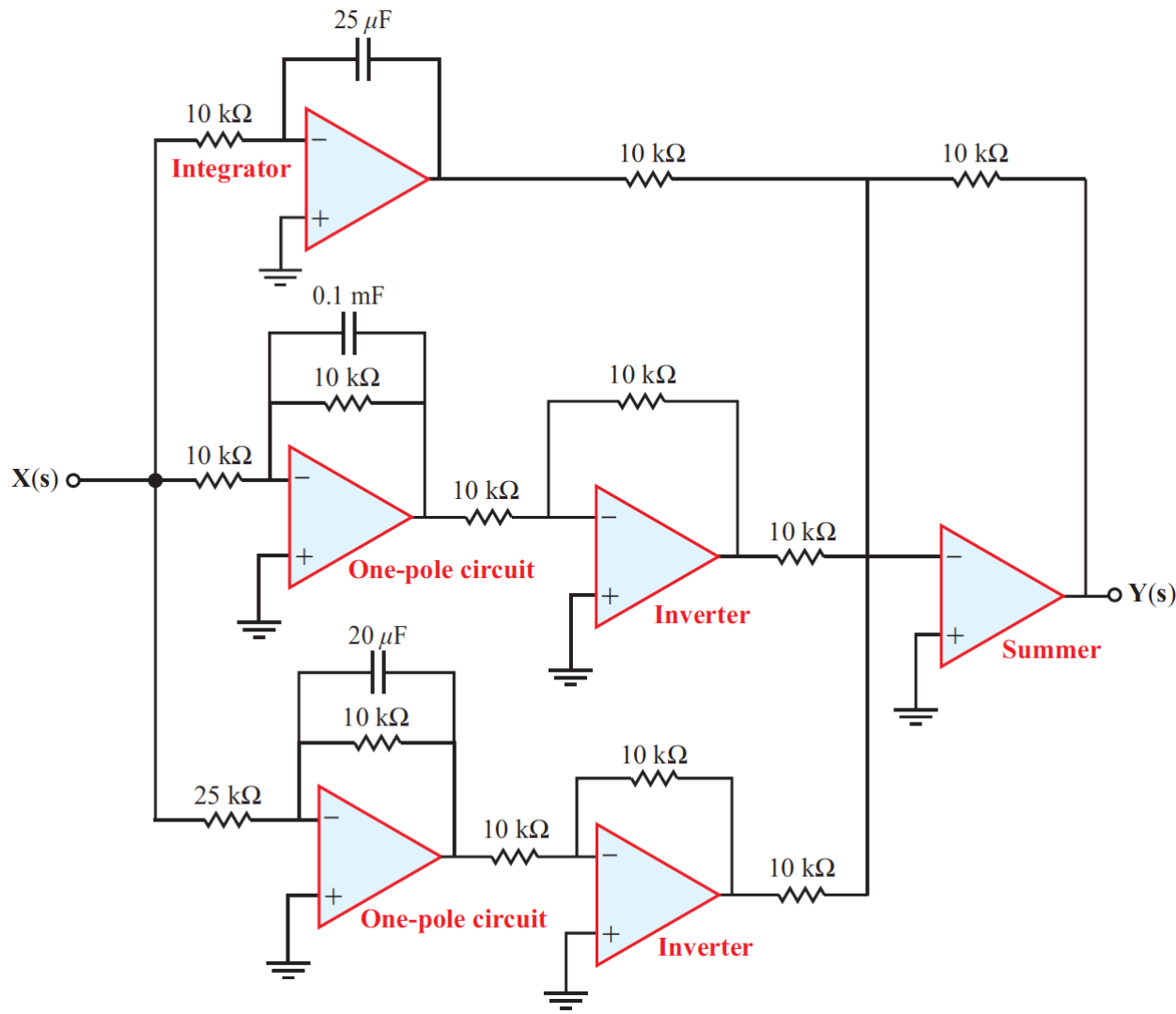
$$Y(s) = H_1(s)E(s)$$

$$= -\frac{2}{s}[-X(s) + 0.2Y(s)]$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{2}{s + 0.4}$$



Activity



Show that this circuit implements the following transfer function

$$H(s) = \frac{s^2 + 17s + 20}{s(s^2 + 6s + 5)}$$

Figure 4-24 of the textbook provides helpful hints.

Suggested steps:

Step 1: Recognize the basic op-amp building block circuits

Step 2: Sketch a block diagram representation from Step 1, and derive a transfer function formula

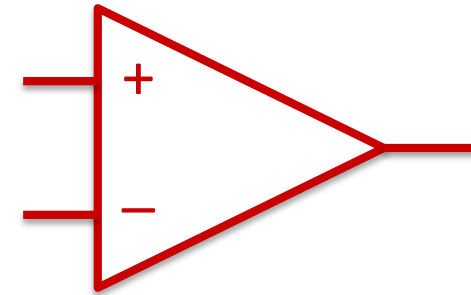
Step 3: Add the partial fractions together



Summary

Today, we learned how to analyze op-amp circuits using Laplace transforms

- Make sure that all op-amps have negative dc feedback
- The two op-amp input voltages are equal
- The op-amp inputs draw **zero** current
- The op-amp output can supply **any** current
- Apply KVL/KCL as usual !



We also learned to build complex systems by connecting simpler systems in a variety of ways

- Cascade, Parallel, Feedback

Important Caveat:

- Our analysis is only valid for the stable case with negative dc feedback



For the Next Class

- Work through the textbook examples in Sections 4-5 & 4-6
- Study Section 4-7 for the next class
- We are skipping 4-8, 4-9, 4-10, 4-11, 4-12

How to Study:

1. Read the assigned sections to understand the basic concepts, and memorize them
2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test
3. Write down your questions and bring them to class